



CONTACT INFORMATION

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Cristina Florena BĂNICĂ

PhD Candidate in Industrial Engineering

National University of Science and Technology POLITEHNICA
Bucharest, Pitești University Centre

Engineer

RATEN ICN Pitești, Mioveni, Romania

EDUCATION

PhD Candidate in Industrial Engineering, 2023 to present

National University of Science and Technology POLITEHNICA Bucharest, Pitești
University Centre

Topic: Contributions to Improving the Functional Design of Parts Produced via 3D
Printing Techniques Using Advanced Analysis and Optimization Methods
Since 21/09/2023

Master's studies in Engineering, 2020 to 2022

University of Pitești, Faculty of Mechanics and Technology
Programme: Product Manufacturing Engineering and Management
05/10/2020 to 22/07/2022

Engineer, 2016 to 2020

University of Pitești, Faculty of Mechanics and Technology
Field: Industrial Economic Engineering
03/09/2016 to 28/07/2020

Natural Sciences, 2012 to 2016

Ion C. Brătianu National College, Pitești
17/08/2012 to 20/06/2016

PROFESSIONAL EXPERIENCE

Engineer (04/10/2021 to present)

RATEN ICN Pitești, Mioveni, Romania

Proposes R&D and technological engineering projects and commercial contracts;
carries out project tasks as project lead. Develops automation and control
solutions for nuclear and industrial applications, CAD designs, technical
documentation, manufacturing technologies and quality plans. Verifies technical
and quality requirements with other departments. Presents research results and
proposes scientific papers.

Economic Engineer (27/02/2021 to 01/10/2021)

S.C. Forum Consult Invest, Pitești, Romania

Assessed client eligibility and documentation. Prepared funding applications,
business plans, financial forecasts, pre-contract documents, visual identity
materials, payment and reimbursement claims, and periodic reports. Advised
clients on eligible expenses, answered clarification requests and contributed to
tender specifications.

Trainee (02/03/2020 to 15/04/2020)

S.C. Automobile Dacia S.A., Mioveni, Romania

Studied gearbox assembly cycle times, analyzed workstations, timed operators'
tasks and proposed process improvements.

Production Manager Assistant (01/07/2019 to 31/10/2019)

Novares Automotive Romania, Mărcăineni, Romania

Reported production data and scrap by defect type. Entered production
reversals, machine production records and Quality Wall records.

RESEARCH INTERESTS

Additive Manufacturing and 3D Printing
Dimensional Accuracy and Quality of Printed Components
Product Design and Manufacturing Technologies
Industrial Management and Process Improvement

SELECTED RESEARCH PROJECTS

Doctoral research in Industrial Engineering, NUST POLITEHNICA Bucharest, Pitești University Centre (since 2023).
Research, development and technological engineering activities at RATEN ICN Pitești: automation and control systems for nuclear and industrial applications (since 2021).

SCIENTIFIC OUTPUT

Co-author of 5 publications listed in the final section (2018 to 2025).
Scientific communications on additive manufacturing, industrial management, mechanics and organizational conflict.
2018, University of Pitești: Design of a Mechanical Gear Reducer Using KissSoft Software (Mechanics and Machine Elements).
2018, SECOSAFT, Sibiu, 19 to 21 April: The Management of an Organizational Conflict (Economic and Legal Sciences).

AWARDS AND DISTINCTIONS

Second Prize, 2022, University of Pitești: Possibilities for Obtaining External Threads by 3D Printing.
First Prize, 2020, University of Pitești: Study on Cycle Time Improvement to Increase Gearbox Assembly Line Performance and Analysis of the Assembly Process.
First Prize, 2019, SECOSAFT, Sibiu: Multisensor Platform for Monitoring Microclimate Parameters in a Room.
Second Prize, 2018, University of Pitești: Managing Conflict in Organizations.
First Prize, 2018, University of Pitești: Study on Conflict Mediation in a Working Group.
First Prize, 2018, Student Scientific Research Session “Prof. Univ. Dr. Eng. Moise Țuțurea”, Brașov: Conflict Mediation in a Working Group.
Honourable Mention, 2017, University of Pitești: On the Equilibrium Equations of a Particle on a Frictionless Surface or Curve.

INTERNATIONAL COLLABORATIONS & MOBILITY

Erasmus mobility, Lodz University of Technology, 01/06/2018 to 31/08/2018.
Laboratory visits; Autodesk Inventor 2018; Mitsubishi robot programming; presentations of industrial robots, a magnetron, a digital microscope and a robotic surgical system; Kia Sportage automatic transmission disassembly and assembly; product design and manufacturing technologies.

TECHNICAL SKILLS

Software: CATIA V5, AutoCAD, Fusion 360, Autodesk Inventor, MATLAB, Microsoft Office

LANGUAGES

Romanian: Mother tongue
English: B2, Cambridge English Certificate (2016)

PERSONAL COMPETENCIES

Analytical, adaptable, detail-oriented
Strong communication and teamwork skills
Organized, proactive and committed to continuous learning

DATE: 2026

CRISTINA FLORENA BĂNICĂ

SELECTED RECENT PUBLICATIONS

- 2025 Rizea, A.-D.; Bănică, C.-F.; Georgescu, T.; Sover, A.; Anghel, D.-C. Dimensional Accuracy Analysis of Splined Shafts and Hubs Obtained by FDM 3D Printing Using a Genetic Algorithm and Artificial Neural Network. *Applied Sciences*.
- 2024 Bănică, C.-F.; Sover, A.; Anghel, D.-C. Printing the Future Layer by Layer: A Comprehensive Exploration of Additive Manufacturing in the Era of Industry 4.0. *Applied Sciences* 14, 9919. <https://doi.org/10.3390/app14219919>
- 2023 Rizea, A.-D.; Anghel, D.-C.; Bănică, C.-F. Contributions to Increasing the Quality of Outer Threads Obtained through 3D Printing. *Acta Technica Napocensis: Applied Mathematics, Mechanics, and Engineering*. WOS:001267255200020.
- 2019 Bănică, C.-F.; Belu, N. Application of 8D Methodology: An Effective Problem Solving Tool in Automotive Industry. *Scientific Bulletin, Automotive Series*, No. 29. Faculty of Mechanics and Technology, University of Pitești, Romania.
- 2018 Woźniak, M.; Bănică, C.-F.; Diaconu, M.; Ozuna, G.; Jóźwiak, P.; Siczek, K. Investigations on Wear and Friction in the SI Engine Valvetrain. *Combustion Engines*.